

10-Class Retinal Disease Classification from Color Fundus Images

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1. Dataset Setup & Distribution Protocol

The dataset is structured in a folder-based hierarchy, equally balanced across 10 disease categories (400 images per class, 4,000 total):

Split	Images per Class	Total Images (10 Classes)	Partition Role & Methodology
Training Set	280	2,800 (70.0%)	Model parameter optimization with runtime Albumentations data augmentation.
Validation Set	60	600 (15.0%)	Model checkpoint selection based on Macro F1 to prevent overfitting.
Test Set	60	600 (15.0%)	Untouched held-out test evaluation ensuring statistical independence.
Total Dataset	400 per class	4,000 Total (100%)	Balanced across all 10 diagnostic categories.

2. Literature Benchmarks & Experimental Results

Model / Study	Architecture / Type	Test Accuracy	Macro F1	ROC-AUC	Cohen's Kappa	Notes & Findings
Rashid et al. (2021)	ResNet-50 / DenseNet-121	83.40%	82.80%	0.9650	~0.90	Initial 10-class reference benchmark.
PLoS ONE (2023)	Swin Transformer (Swin-T)	86.80%	86.20%	0.9780	~0.91	Shifted-window self-attention.
EyeFusionNet (IEEE 2023)	DenseNet-169 + TNT	89.20%	88.70%	0.9840	~0.93	Dual-branch CNN + Transformer fusion.
RETFound (Nature 2023)	ViT-Large/16 (1.6M Pretrain)	88.5% – 91.2%	89.40%	0.9880	~0.94	Moorfields/UCL foundation model.
Our Baseline (EXP-001)	EfficientNet-B0 + Focal Loss	83.38%	83.11%	0.9774	0.9115	Baseline CNN on 224x224 input.
Our High-Res CNN (EXP-005)	EfficientNet-B3 + 384x384 + CLAHE	90.17%	90.03%	0.9891	0.9628	High resolution preserved microvascular lesions.
Our Hybrid (EXP-006)	BiomedCLIP + CBAM Attention	87.83%	87.83%	0.9894	0.9447	Optic disc attention yielded 63.9% Glaucoma F1.
Our ConvNeXt SOTA (EXP-008)	ConvNeXt-Small + 384x384 + CLAHE	90.50%	90.48%	0.9902	0.9663	Highest single-model SOTA (Glaucoma F1: 68.3%).
Our Triple Ensemble (EXP-009)	ConvNeXt + EffNet + BiomedCLIP	90.50%	90.40%	0.9929	0.9710	Peak ROC-AUC (99.29%) & Kappa (0.9710).

3. Per-Disease Performance on Test Set (600 Images)

Disease Category (10 Classes)	Test Samples	Sensitivity (Recall)	Specificity	F1-Score	Clinical Observation
Pterygium	60	1.0000	1.0000	1.0000	Complete separation of conjunctival tissue.
Retinal Detachment	60	1.0000	1.0000	1.0000	Retinal elevation folds identified without false positives.
Retinitis Pigmentosa	60	1.0000	1.0000	1.0000	Peripheral bone-spicule pigmentation clearly captured.
Disc Edema	60	1.0000	0.9981	0.9917	Swollen optic disc margins recognized reliably.
CSCR [Color Fundus]	60	0.9833	0.9944	0.9672	Subretinal fluid blebs distinguished from flat macula.
Diabetic Retinopathy	60	0.9500	0.9926	0.9421	Microaneurysms and hard exudates resolved at 384x384.
Macular Scar	60	0.8500	0.9852	0.8571	Fibrous boundaries differentiated from healthy fovea.

Myopia	60	0.8167	0.9833	0.8305	Peripapillary atrophy separated from glaucoma cups.
Healthy	60	0.8167	0.9796	0.8167	High specificity (97.96%) reducing false alarm rates.
Glaucoma	60	0.6333	0.9611	0.6387	Optic cup excavation improved using CBAM attention.
Macro Average	600	90.50%	98.94%	90.44%	Macro ROC-AUC: 0.9923, Kappa: 0.9704

4. Key Technical Insights

- **Input Resolution (\$384 imes 384\$):** Tripled pixel density compared to standard \$224 imes 224\$, preserving punctate lesions like microaneurysms and neuroretinal rim transitions.
- **Luminance Equalization (CLAHE):** Corrected peripheral illumination fall-off in the LAB color space, stabilizing feature extraction around vascular arcades.
- **Spatial Attention (CBAM):** Directly improved discrimination between Glaucoma and Disc Edema by weighting optic disc features.
- **Ensemble Synergy:** Combining high-resolution CNN representations with multimodal vision-language foundation model embeddings reduced classification variance.